understand it—is determined by the first. It is an entirely reasonable objection, just so long as physics reassures us that the external flow of time is real, universal, and in keeping with our intuitions. But if physics shows us instead that such time is *not* an elementary part of reality, can we continue to overlook Augustine's observation and treat it as irrelevant to the true nature of time?

Inquiry into perception of the *internal* rather than *external* nature of time recurs frequently in Western philosophy. Kant discusses the nature of space and time in his *Critique of Pure Reason*, and interprets both space and time as a priori forms of knowledge—that is to say, things that do not just relate to the objective world but also to the way in which a subject apprehends it. But he also observes that whereas space is shaped by our *external* sense, that is to say, by our way of ordering things that we see in the world *outside* of us, time is shaped by our *internal* sense, that is to say, by our way of ordering internal states *within* ourselves. Once again: the basis of the temporal structure of the world is to be sought in something that closely relates to our way of thinking and perceiving, to our consciousness. This remains true without having to get tangled up in Kantian transcendentalism.

Husserl reprises Augustine when he describes the shaping of experience in terms of "retention"—using, like him, the metaphor of listening to a melody[113] (the world, in the meantime, has become bourgeois, with melodies replacing hymns): in the moment that we hear a note, the previous note is "retained," then that one also becomes part of the retention—and so on, running them together in such a way that the present contains a continuous trace of the past, becoming gradually more blurred.[114] It is by way of this process of retention, according to Husserl, that phenomena "constitute time." The diagram on the following page is Husserl's: the horizontal axis from A to E represents time passing; the vertical axis from E to A' represents the "retention" of moment E, where the progressive subsidence leads from A to A'. Phenomena constitute time because, at the moment, E, P', and A' exist. The interesting point here is that the source of the phenomenology of time is not identified by Husserl in the hypothetical objective succession of phenomena (the horizontal line) but rather in memory (similar to anticipation, called "protention" by Husserl), that is to say, by the vertical line in the diagram. My point is that this continues to be valid (in a natural philosophy) even in a physical world where there is no physical time globally organized in a linear way but only traces generated by varying entropy.